LAKSHYA

JEE 2027

Lecture No.- 09 | Mathematics

Determinants

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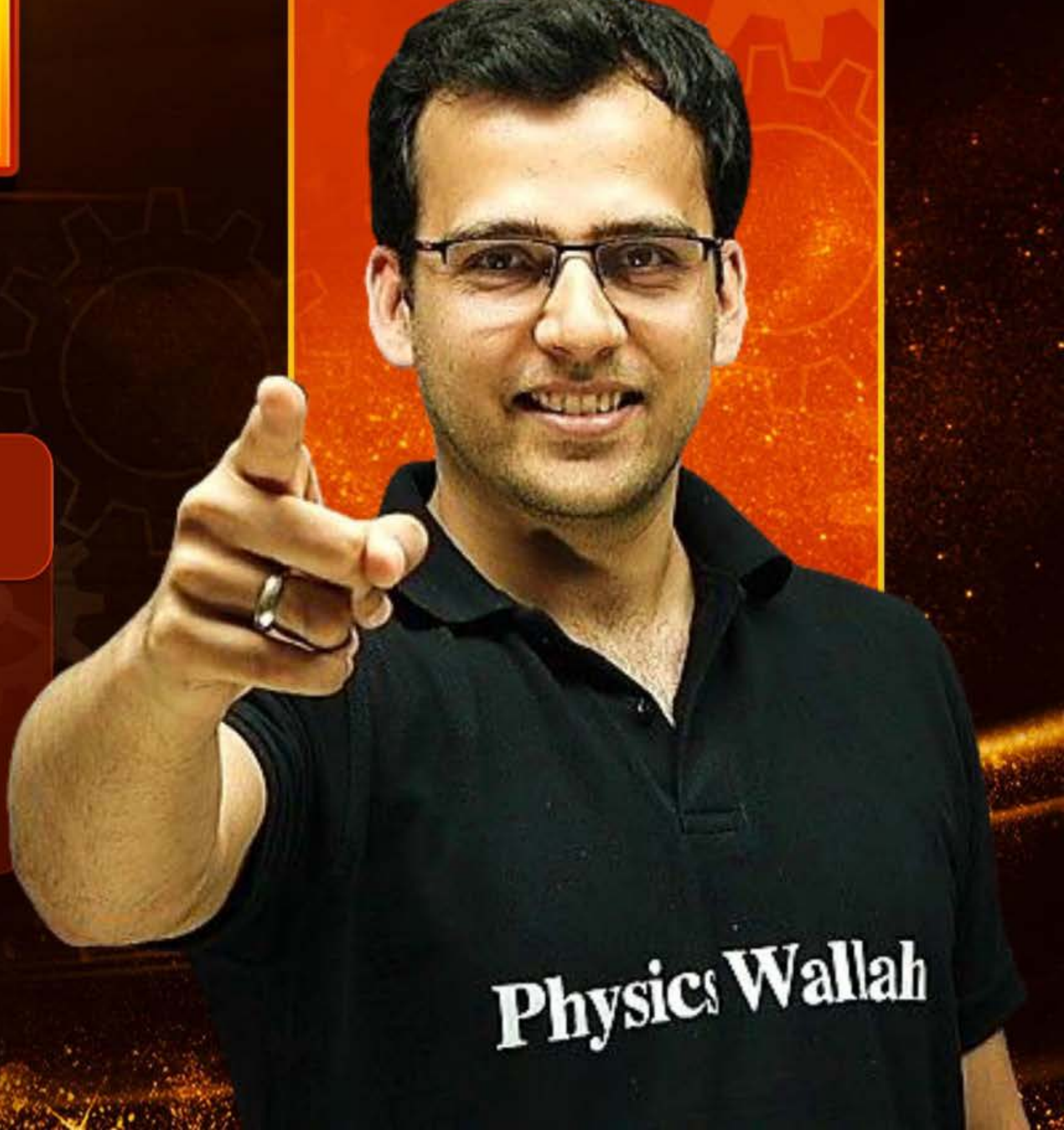

Topics *to be covered*

1. HW Discussion
2. Differentiation of Det Contd..
3. Cramer's Rule 70% PYQS (Mains)

RECAP Differentiation of Determinant

Differentiate one row or one column at a time (Just like Product Rule)
Example-

If $D = \begin{vmatrix} f(x) & g(x) & h(x) \\ p(x) & q(x) & r(x) \\ l(x) & m(x) & n(x) \end{vmatrix}$ then

$$D' = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ p(x) & q(x) & r(x) \\ l(x) & m(x) & n(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ p'(x) & q'(x) & r'(x) \\ l(x) & m(x) & n(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ p(x) & q(x) & r(x) \\ l'(x) & m'(x) & n'(x) \end{vmatrix}$$

HW Question [JEE Main 2025 (April)]

[Ans. A]

If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}$, $x \in \mathbb{R}$, then $\frac{d^2y}{dx^2} + y$ is equal to: $\rightarrow$ (1)

(A) -1

(B) 28

(C) 27

(D) 1

$\boxed{M-1}$

$$\frac{dy}{dx} = \begin{vmatrix} \cos x & -\sin x & \cos x - \sin x \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix} + \underbrace{\begin{vmatrix} 0 & 0 & 0 \\ & & \\ & & \end{vmatrix}}_{0} + \underbrace{\begin{vmatrix} & & \\ & & \\ 0 & 0 & 0 \end{vmatrix}}_{0}$$

$$\frac{d^2y}{dx^2} = \begin{vmatrix} -\sin x & -\cos x & -\sin x - \cos x \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix} \rightarrow (2)$$

(1) + (2) (P-6 Converse)

$$\frac{d^2y}{dx^2} + y = \begin{vmatrix} 0 & 0 & 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix} = 27 - 28 = -1$$

HW Question [JEE Main 2025 (April)]

[Ans. A]

If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}, x \in \mathbb{R}$, then $\frac{d^2y}{dx^2} + y$ is equal to:

(A) −1 ✓

(B) 28

(C) 27

(D) 1

$\boxed{M\text{-}2}$ $C_1 \to C_1 - C_3$

$$\begin{vmatrix} -(\cos x + 1) & \cos x & \sin x + \cos x + 1 \\ 0 & 28 & 27 \\ 0 & 1 & 1 \end{vmatrix}$$

$-(\cos x + 1)[28 - 27]$

$\boxed{y(x) = -\cos x - 1}$ ✓ $\longrightarrow$ ①

$\frac{dy}{dx} = \sin x$

$\frac{d^2y}{dx^2} = \cos x \longrightarrow$ ②

① + ②

$\frac{d^2y}{dx^2} + y = -1$

Question [JEE Mains 2024]

[Ans. A]

If $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$ then $\frac{1}{5}f'(0)$ is equal to ____ 0

(A) 0 ✓

(B) 1

(C) 2

(D) 6

$R_1 \to R_1 - R_2$ & $R_2 \to R_2 - R_3$

$$\begin{vmatrix} -3 & 0 & 3 \\ 3 & -3 & 0 \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$$

$$9\begin{vmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$$

$-1[-\sin^2 2x] + 1[3+2\sin^4 x + 2\cos^4 x]$

$\sin^2 2x + 3 + 2(\underbrace{\sin^4 x + \cos^4 x})$

$\sin^2 2x + 3 + 2(1 - 2\sin^2 x\cos^2 x)$

$9[\cancel{\sin^2 2x} + 5 - \cancel{4\sin^2 x\cos^2 x}]$

$\boxed{f(x) = 45}$

$f'(x) = 0$

$f'(0) = 0$

$\boxed{M-2}$ ✓

$$f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$$

$$\frac{d}{dx}(\cos x)^4 = 4(\cos x)^3(-\sin x)$$

Diff karke x ko zero

$$\frac{d}{dx}(\sin x)^4 = 4(\underbrace{\sin x}) ^3 \cos x$$

$$f'(0) = \begin{vmatrix} 0 & - & - \\ 0 & - & - \\ 0 & - & - \end{vmatrix} + \begin{vmatrix} - & 0 & - \\ - & 0 & - \\ - & 0 & - \end{vmatrix}$$

$$+ \begin{vmatrix} - & - & 0 \\ - & - & 0 \\ - & - & 0 \end{vmatrix}$$

QUESTION [JEE MAIN 2023]

[Ans. 6]

Let $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$.

If $\sum_{k=1}^{n} D_k = 96$, then n is equal to

$\sum 2K = 2 \sum K$

$= \cancel{2} \frac{n(n+1)}{\cancel{2}}$

$1+3+5+ \ldots + 2n-1 = n^2$

$$\sum_{K=1}^{n} D_K = \begin{vmatrix} \sum_{K=1}^{n} 1 & \sum_{K=1}^{n} 2K & \sum_{K=1}^{n} 2K-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$$

$$\begin{vmatrix} n & n(n+1) & n^2 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix} = 96$$

$R_1 \to R_1 - R_2$

$$\begin{vmatrix} 0 & -2 & 0 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$$

$2\left[n(n^2+n+2) - n^3\right]$

$2n\left[\cancel{n^2}+n+2-\cancel{n^2}\right]$

$2n(n+2) = 96.$

$n(n+2) = 48 = 6 \times 8$

$\boxed{n=6}$

Question [JEE Mains 2024]

[Ans. C]

If $f(x)=\begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix}$ for all $x \in \mathbb{R}$, then $2f(0)+f'(0)$ is equal to:

(A) 48

(B) 24

(C) 42

(D) 18

$$f(0)=\begin{vmatrix} 0 & 1 & 1 \\ 2 & 0 & 6 \\ 0 & 4 & -2 \end{vmatrix} = -2[-2-4] = 12$$

$$\boxed{f'(0)=24-6=18}$$ ✓

$$f'(x)=\begin{vmatrix} 3x^2 & 4x & 3 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{vmatrix} + \begin{vmatrix} & & \\ 6x & 2 & 3x^2 \\ & & \end{vmatrix} + \begin{vmatrix} & & \\ & & \\ 3x^2-1 & 0 & 2x \end{vmatrix}$$

$$f'(0)=\underbrace{\begin{vmatrix} 0 & 0 & 3 \\ 2 & 0 & 6 \\ 0 & 4 & -2 \end{vmatrix}}_{24} + \begin{vmatrix} 0 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 4 & -2 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 1 \\ 2 & 0 & 6 \\ -1 & 0 & 0 \end{vmatrix} = -6$$

Question [JEE Mains 2026 (23 Jan., Shift 1)]

[Ans. [redacted]

Among the statements:

I: If $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$, and

Wrong

II : If $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = \mathrm{p}x + q$, then $p^2 = 196q^2$,

(A) Only II is true

(B) Both are false

(C) Both are true

(D) Only I is true

(1) $1[1-\cos^2\gamma] - \cos\alpha[\cos\alpha - \cos\beta\cos\gamma] + \cos\beta(\cos\alpha\cos\gamma - \cos\beta]$

$\boxed{\underline{LHS} = 1 - \cos^2\gamma - \cos^2\alpha - \cos^2\beta + 2\cos\alpha\cos\beta\cos\gamma}$

RHS: $-\cos\alpha[-\cos\beta\cos\gamma] + \cos\beta[\cos\alpha\cos\gamma]$

$\Rightarrow \boxed{RHS = 2\cos\alpha\cos\beta\cos\gamma}$

$1 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma = 0 \Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Question [JEE Mains 2026 (23 Jan., Shift 1)]

[Ans. B]

Among the statements:

I: If $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$, and

II : If $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = px + q$, then $p^2 = 196q^2$,

(A) Only II is true

(B) Both are false

(C) Both are true

(D) Only I is true

$2x-1+x-2$

$3x-3$

Wrong

$(24)^2 = 196(12)^2$ $\quad p = 24$

$q = -12$

$R_1 \to R_1 + R_3 - R_2$

$\begin{vmatrix} 4 & 0 & 0 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = 4\left[3x(2x-1) - (2x-1)(3x-3)\right]$

$4(2x-1)\left[\cancel{3x} - (\cancel{3x}-3)\right]$

$4(2x-1)\cdot 3$

$= 12(2x-1)$

$= 24x-12$

HW Question

Put $x=0$ for a_7

[Ans. A, C]

If x is real and $\Delta(x)=\begin{vmatrix} x^2+x & 2x-1 & x+3 \\ 3x+1 & x^2+2 & x^3-3 \\ x-3 & x^2+4 & 2x \end{vmatrix}=a_0x^7+a_1x^6+a_2x^5+\cdots+a_6x+a_7$, then:

Put $x=1$

$$\begin{vmatrix} 2 & 1 & 4 \\ 4 & 3 & -2 \\ -2 & 5 & 2 \end{vmatrix}=\underbrace{a_0+a_1+a_2+\cdots a_6}+21$$

(A) $a_7=21$ ⟶ Put $x=0$

(B) $\Delta(-1)=32$

(C) $\sum_{k=0}^{6} a_k=111$ $a_0+a_1+a_2+a_3+a_4+a_5+a_6=?$

(D) $\Delta(1)=121$

Brain Teaser

[Ans $a_5 = 34$, [redacted]

If x is real and $\Delta(x) = \begin{vmatrix} x^2+x & 2x-1 & x+3 \\ 3x+1 & x^2+2 & x^3-3 \\ x-3 & x^2+4 & 2x \end{vmatrix} = a_0x^7 + a_1x^6 + a_2x^5 + \cdots + a_6x + a_7$, then also find a_5, a_0 & a_1

$$a_6 = f'(0)$$

$$\Delta(x) = a_0x^7 + a_1x^6 + \dots a_5x^2 + a_6x + a_7$$

$$\Delta'(x) = 7a_0x^6 + 6a_1x^5 + \dots 2a_5x + a_6$$

$$\Delta''(x) = 42a_0x^5 + 30a_1x^4 + \dots 2a_5 + 0$$

$$\boxed{x=0}$$

$$\Delta''(0) = 2a_5$$

$$\boxed{a_5 = \frac{\Delta''(0)}{2}}$$

HW

Brain Teaser

[Ans $a_5 = 34$, $a_0 = -1$ & $a_1 = -1$

If x is real and $\Delta(x) = \begin{vmatrix} x^2+x & 2x-1 & x+3 \\ 3x+1 & x^2+2 & x^3-3 \\ x-3 & x^2+4 & 2x \end{vmatrix} = \boxed{a_0x^7 + a_1x^6 + a_2x^5 + \cdots + a_6x + a_7,}$ then also find a_5, a_0 & a_1

For a_0 & a_1 replace $x \to 1/x$

$$\begin{array}{c} x^2 \\ x^3 \\ x^2 \end{array}\begin{vmatrix} \frac{1}{x^2}+\frac{1}{x} & \frac{2}{x}-1 & \frac{1}{x}+3 \\ \frac{3}{x}+1 & \frac{1}{x^2}+2 & \frac{1}{x^3}-3 \\ \frac{1}{x}-3 & \frac{1}{x^2}+4 & \frac{2}{x} \end{vmatrix} = \frac{a_0}{x^7}+\frac{a_1}{x^6}+\frac{a_2}{x^5}+\cdots\frac{a_6}{x}+a_7$$

x^7

$$\begin{vmatrix} 1+x & 2x-x^2 & x+3x^2 \\ 3x^2+x^3 & x+2x^3 & 1-3x^3 \\ x-3x^2 & 1+4x^2 & 2x \end{vmatrix} = a_0+a_1x+a_2x^2+\cdots a_6x^6+a_7x^7$$

$\boxed{\text{Put } x=0}$ ✓

Put $x=0$

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = a_0$$

$$\boxed{-1 = a_0}$$

for a_1 diff wrt x & put $\boxed{x=0}$

HW

Brain Teaser

M-2

[Ans $a_5 = 34$, $a_0 = -1$ & $a_1 = -1$

If x is real and $\Delta(x) = \begin{vmatrix} x^2+x & 2x-1 & x+3 \\ 3x+1 & x^2+2 & x^3-3 \\ x-3 & x^2+4 & 2x \end{vmatrix} = \boxed{a_0 x^7 + a_1 x^6 + a_2 x^5 + \cdots + a_6 x + a_7}$, then also find a_5, a_0 & a_1

$$\cancel{x^3}\,\cancel{x^2}\,\cancel{x}\begin{vmatrix} 1+\frac{1}{x} & \frac{2}{x}-\frac{1}{x^2} & \frac{1}{x}+\frac{3}{x^2} \\ \frac{3}{x^2}+\frac{1}{x^3} & \frac{1}{x}+\frac{2}{x^3} & 1-\frac{3}{x^3} \\ \frac{1}{x}-\frac{3}{x^2} & 1+\frac{4}{x^2} & \frac{2}{x} \end{vmatrix} = \cancel{x^7}\left[a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_6}{x^6} + \frac{a_7}{x^7}\right]$$

$\boxed{x \to \infty}$

$\frac{1}{x} \to 0$

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = a_0$$

Important Note

A function $f(x)$ *can be expressed as*

$$\boxed{f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + +a_4 x^4 \dots\dots\dots\dots\dots}$$

Where a_0 = $f(0)$

Where a_1 = $f'(0)$

Where a_2 = $\frac{f''(0)}{2!}$

Where a_3 = $\frac{f'''(0)}{3!}$

$f'(x) = a_1 + 2a_2x + 3a_3x^2 \dots$

$f''(x) = \underbrace{2a_2} + 6a_3x + \dots$

$f''(0) = 2a_2$

$a_2 = \frac{f''(0)}{2}$

$f'''(0) = 6a_3$

$a_3 = \frac{f'''(0)}{6}$

$\boxed{\text{Maclaurin Series}}$ $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f''''(0)}{4!}x^4 \dots\dots\dots\dots\dots \infty$

In $\underline{\underline{\text{Calculus}}}$

$f(x) = x^4$

$f'(x) = 4x^3$

$f''(x) = 4 \cdot 3x^2$

$f'''(x) = 4 \cdot 3 \cdot 2x^1$

Derivation of Expansion of e^x

$$\boxed{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} \ldots\ldots\ldots\ldots\ldots \infty}$$

Applications

Proof

Maclaurin Series $\boxed{f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f''''(0)}{4!}x^4 \ldots\ldots\ldots\ldots \infty}$

$f(x) = e^x$

$f(0) = 1$

$f'(x) = e^x \Rightarrow f'(0) = 1$

$f''(0) = e^x = e^0 = 1$

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \ldots \infty$$

Ex →

$f(x) = \sin x \Rightarrow f'(x) = \cos x$

$f(0) = 0$

$f'(0) = 1$

$f''(x) = -\sin x$

$f''(0) = 0$

Linear Equations in Two Variables

$$\boxed{\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned}}$$

Unique Sol^n ✓

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

No Sol^n

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Infinite Sol^n

$$a_1/a_2 = b_1/b_2 = c_1/c_2$$

→ Consistent

Ex →

$3x + 5y = 7$

$6x + 10y = 11$

$$\frac{3}{6} = \frac{5}{10} \neq \frac{7}{11}$$

$\Rightarrow$ No sol^n

Linear Equations in Two Variables

$$a_1x + b_1y = c_1$$
$$a_2x + b_2y = c_2$$

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

$$D_1 = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$$

$$D_2 = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

$$x = \frac{D_1}{D}, \quad y = \frac{D_2}{D}$$

Cross Multiplication method

Nature of Sol. Linear Equations in Two Var.

$a_1x + b_1y = c_1$

$a_2x + b_2y = c_2$

Linear Equations in 2 Variables

- **Consistent**
  - Intersecting Lines: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ — Unique solution
  - Coincident Lines: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ — Infinite solution
- **Inconsistent**
  - Parallel Lines: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ — No solution

Question [JEE Main 2013]

[Ans. B]

The number of values of k for which the system of linear equations

$$\boxed{\begin{aligned}(k+1)x + 8y &= 4k,\\ kx + (k+3)y &= 3k - 1.\end{aligned}}$$

has no solution is ?

$$\frac{a_1}{a_2} = \boxed{\frac{b_1}{b_2} \neq \frac{c_1}{c_2}}$$

(A) Infinite

(B) 1 ✓

(C) 2

(D) 3

$$\frac{k+1}{k} = \frac{8}{k+3}$$

$$(k+1)(k+3) = 8k$$

$$k^2 + 4k + 3 = 8k$$

$$k^2 - 4k + 3 = 0$$

$$(k-3)(k-1) = 0$$

$$k = 3, 1$$

Also: $\frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

$$\frac{8}{k+3} \neq \frac{4k}{3k-1}$$

Put $k = 1$

$$\frac{8}{4} \neq \frac{4}{2} \Rightarrow 2 \neq 2$$

False

$k = 1$ Reject

Put $\boxed{k = 3}$

$$\frac{8}{6} \neq \frac{12}{8} \Rightarrow \text{True.}$$

$k = 3$ is accepted

Question [JEE Adv. 2016]

HW

[Ans. B,C,D]

Let $a, \lambda, \mu \in R$. Consider the system of linear equations

$$ax + 2y = \lambda,$$
$$3x - 2y = \mu.$$

Which of the following statement(s) is(are) correct?

(A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ

(B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

Solving System of Linear Equation in 3 Variable

Case-1: Non-homogenous system of equation:

$$a_1x + b_1y + c_1z = d_1$$ → zero degree
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

Case-2: Homogenous system of equation: same degree

"all constants are zero"

$$a_1x + b_1y + c_1z = 0$$
$$a_2x + b_2y + c_2z = 0$$
$$a_3x + b_3y + c_3z = 0$$

Ex →

$2x + 3y + 4z = 0$

$3x - 2y + 7z = 5$

Cramer's Rule for Non-Homogeneous System of Equations

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

$$Consider\ D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

then $x = \frac{D_1}{D}$, $y = \frac{D_2}{D}$, $z = \frac{D_3}{D}$

Proof: $D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$ $\Rightarrow \boxed{x = \frac{D_1}{D}}$ ✓

$$D_1 = \begin{vmatrix} a_1x+b_1y+c_1z & b_1 & c_1 \\ a_2x+b_2y+c_2z & b_2 & c_2 \\ a_3x+b_3y+c_3z & b_3 & c_3 \end{vmatrix}$$

Similarly

$D_2 = yD$

$D_3 = zD$

$$C_1 \longrightarrow C_1 - yC_2 - zC_3$$

$$D_1 = x\underbrace{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}_{D}$$

$D_1 = xD$

Question

[Ans x = 1, y = 2, z = 3

Using Determinants, Solve the following system of equations for x , y & z

$$\boxed{\begin{aligned} x+y+z &= 6, \\ y+3z &= 11 \\ x-2y+z &= 0 \end{aligned}}$$

$$\left.\begin{aligned} x+y+z &= 6 \\ 0x+y+3z &= 11 \\ x-2y+z &= 0 \end{aligned}\right\}$$

$x = \frac{D_1}{D} = \frac{9}{9} = 1$

$y = \frac{D_2}{D} = \frac{18}{9} = 2$

$z = 3$

$$D_3 = \begin{vmatrix} 1 & 1 & 6 \\ 0 & 1 & 11 \\ 1 & -2 & 0 \end{vmatrix} = 27$$

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{vmatrix} = 1[1+6]-1[-3]+1[-1]$$

$= 7+3-1 = 9$

$$D_1 = \begin{vmatrix} 6 & 1 & 1 \\ 11 & 1 & 3 \\ 0 & -2 & 1 \end{vmatrix} = 6[1+6]-1[11]+1[-22]$$

$42-22-11 = 20-11 = 9$

$$D_2 = \begin{vmatrix} 1 & 6 & 1 \\ 0 & 11 & 3 \\ 1 & 0 & 1 \end{vmatrix} = 1[11]-6[-3]+1[-11]$$

$= 18$

Nature of Solutions

Consistent

Case-1

If $D \neq 0$ & $D_1 = D_2 = D_3 = 0$ then the system of equation has Unique trivial solution

$\Rightarrow \underbrace{x = 0,\ y = 0,\ z = 0}$ ✓

(teeno zero)

1.

Consistent

Case-2

If $D \neq 0$ & at-least one of D_1, D_2 or $D_3 \neq 0$ then the system has Unique non trivial sol^n

$D = 5,\ D_1 = 10,\ D_2 = 0,\ D_3 = 0$

$x = 2,\ y = 0,\ z = 0$

Inconsistent

Case-3

If D = 0 & at-least one of D_1, D_2 or $D_3 \neq 0$ then the system has no solution

$D = 0,\ D_1 = 2,\ D_3 = 0,\ D_2 = 5$ → not possible

Consistent

Case-4

If $D = 0$ & $D_1 = D_2 = D_3 = 0$ then the system has infinite sol^n.

Question [JEE Mains 2025]

[Ans. 26]

If the system of equation:

$$2x + \lambda y + 3z = 5$$
$$3x + 2y - z = 7$$
$$4x + 5y + \mu z = 9$$

has infinitely many solutions, then $(\lambda^2 + \mu^2)$ is equal to ______. $(-1)^2 + (-5)^2 = 26$

$$\boxed{D = D_1 = D_2 = D_3 = 0}$$

$\lambda = -1$

$$D = \begin{vmatrix} 2 & -1 & 3 \\ 3 & 2 & -1 \\ 4 & 5 & \mu \end{vmatrix} = 0$$

$$2[2\mu+5] + 1[3\mu+4] + 3[15-8] = 0$$
$$4\mu + 10 + 3\mu + 4 + 21 = 0$$
$$7\mu + 35 = 0 \quad \mu = -5$$

$$D_3 = \begin{vmatrix} 2 & \lambda & 5 \\ 3 & 2 & 7 \\ 4 & 5 & 9 \end{vmatrix} = 0$$

$$2(18-35) - \lambda[27-28] + 5(15-8) = 0$$
$$-17\times 2 + \lambda + 5\times 7 = 0$$
$$-34 + \lambda + 35 = 0$$
$$\boxed{\lambda = -1}$$

Question [JEE Mains 2025 (24 Jan. Shift 1)]

[Ans. A]

If the system of equations $2x - y + z = 4, 5x + \lambda y + 3z = 12, 100x - 47y + \mu z = 212$, has infinitely many solutions, then $\mu - 2\lambda$ is equal to:

(A) 57

(B) 59

(C) 55

(D) 56

Question [JEE Mains 2025]

[Ans. B]

If the system of linear equations $3x + y + \beta z = 3$; $2x + \alpha y - z = -3$; $x + 2y + z = 4$ has infinitely many solutions, then the value of $22\beta - 9\alpha$ is:

(A) 49

(B) 31

(C) 43

(D) 37

Question [JEE Mains 2023]

[Ans. C]

For $\alpha, \beta \in \mathbb{R}$, suppose the system of linear equations

$$x - y + z = 5$$
$$2x + 2y + \alpha z = 8$$
$$3x - y + 4z = \beta$$

has infinitely many solutions. Then α and β are the roots of:

(A) $x^2 - 10x + 16 = 0$

(B) $x^2 + 18x + 56 = 0$

(C) $x^2 - 18x + 56 = 0$

(D) $x^2 - 14x + 24 = 0$

Question [JEE Mains 2024 (Feb.-I)]

[Ans. B]

If the system of equations:

$$2x + 3y - z = 5$$
$$x + \alpha y + 3z = -4$$
$$3x - y + \beta z = 7$$

has infinitely many solutions, then $13\alpha\beta =$

(A) 1110

(B) 1120

(C) 1210

(D) 1220

Question [JEE Mains 2024 (Jan.-II)]

[Ans. 113]

Let for any three distinct consecutive terms a, b, c of an A.P., the lines $ax + by + c = 0$ be concurrent at the point P and $Q(\alpha, \beta)$ be a point such that the system of equations

$x + y + z = 6$

$2x + 5y + \alpha z = \beta$ and

SL

$x + 2y + 3z = 4,$

has infinitely many solutions. Then $(PQ)^2$ is equal to ________.

Homework

HW Module-

TOPIC WISE → Q→11 to 35

THANK

You

 # Target JEE Advanced 2027 !